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PROFESSIONAL SUMMARY

Embedded & Wireless Communication Engineer with 9+ years of experience in embedded systems, RF modem development, 5G NR/LTE Layer-1 software, DSP algorithms, RF calibration, thermal mitigation, power optimization, and modem scheduling. Experienced in Qualcomm modem platforms, LTE/NB-IoT systems, and UHF Radio Modem development for Indian Railways KAVACH systems. Strong expertise in real-time embedded systems, RF performance tuning, debugging, validation, and cross-functional technical leadership.

PROFESSIONAL EXPERIENCE

Sr. Lead Engineer – IINVSYS Pvt. Ltd., Puducherry (Mar 2025 – Present)

- Leading development of UHF Radio Modem for Indian Railways KAVACH safety systems as per RDSO specifications.
- Worked extensively on TI CC1311P3 chipset-based UHF radio modem including RF driver development, SDK integration, board bring-up, RF calibration, RF validation, and hardware-level debugging support.
- Designed and developed UHF controller firmware with complete hardware-software integration.
- Developed RDSO-compliant firmware on STM32 and TI microcontrollers.
- Implemented and optimized DSP algorithms for interoperability, sensitivity, and RF performance improvements.
- Conducted RF troubleshooting, calibration, and performance fine-tuning across multiple operating conditions.
- Designed and implemented OTA firmware upgrade mechanisms, SPI/UART/RS232/I2C interfaces, and production test benches.
- Conducted thermal validation and enabled thermal, current, and voltage protection frameworks for UHF systems.
- Worked closely with hardware, RF, validation, and customer integration teams for debugging and deployment support.

Lead Engineer (Senior) – Qualcomm India Pvt. Ltd., Hyderabad (Jan 2020 – Mar 2025)

- Responsible for Layer-1 scheduler architecture and coordination across Search, DL, UL, Beam Management, and Qualcomm proprietary modem modules.
- Designed and developed Layer-1 Control (L1C) scheduler enhancements for real-time coordination and resource management.
- Worked on 5G NR utilities including Measurement Gap coordination, AFC, thermal mitigation algorithms, and coexistence module maintenance/debugging.
- Designed and developed Sleep Controller features and collaborated with Power Management teams for Micro Sleep, Fast Sleep, Light Sleep, Deep Sleep, and adaptive power-saving techniques.
- Contributed to SAR exposure control and Smart TX feature development to meet FCC compliance requirements.
- Developed and validated thermal mitigation algorithms under varying workloads and RF conditions.
- Worked extensively on memory optimization, code optimization, Klocwork issue fixing, and modem stability improvements.
- Resolved critical modem issues including memory leaks, crashes, hard faults, race conditions, and scheduler deadlocks.

- Mentored 2–3 engineers on modem architecture, debugging, and feature development.

Development Engineer – Mbit Wireless Pvt. Ltd., Chennai (Aug 2016 – Jan 2020)

- Worked on LTE Cat-9, Cat-4/1, Cat-M1, and NB-IoT technologies.
- Developed RF drivers and integrated RF front-end modules with baseband systems.
- Implemented MIPI RFFE interfaces for PA, LNA, and RF switch control.
- Implemented Layer-1 Cell Search algorithms including PSS, SSS, and Frequency/Frame Estimation.
- Contributed to UL PHY modules including Power Control, CSI, PUCCH, PUSCH, and HARQ.
- Performed RF calibration, RF conformance testing, board bring-up, and silicon readiness activities.
- Contributed to power optimization features such as clock gating, power domain management, and dynamic clock scaling.

TECHNICAL SKILLS

Programming: C, Embedded C, Python Basics, C++ Basics, Data Structures

Wireless: 5G NR Layer-1, LTE, Cat-M1, NB-IoT, UHF Modem, DSP Algorithms, RF Calibration, Thermal Mitigation, Beam Management, Smart TX

Embedded Systems: STM32, TI CC1311P3, RF Driver Development, OTA Upgrade, Real-Time Embedded Systems

Interfaces: SPI, UART, RS232, I2C, AXI/AHB, MIPI RFFE

Power Optimization: Micro Sleep, Fast Sleep, Light Sleep, Deep Sleep, Dynamic Clock Scaling

Lab Instruments: Signal Analyzer, Spectrum Analyzer, Anritsu RF Test Equipment, Rohde & Schwarz CMW Platform, Aeroflex RF Test Systems, Logic Analyzer, GPIO timing-level debugging and signal probing

Compliance: RDSO, FCC, RF Conformance, GCF, 3GPP

EDUCATION

B.E – Computer Science and Engineering, Jeppiaar Engineering College, Chennai